

Solution Requirements

Date	01 October 2026
Team ID	6ab7f6f8c23107aaaa72871e
Project Name	FitBuddy - AI Fitness Plan Generator using Gemini Models
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements

The requirements distinguish what FitBuddy does from the quality attributes expected of the solution.

Functional Requirements (FR)

FR-1 Authentication No login service is documented; user_id identifies stored records.	FR-2 Authorization /view-all-users provides an admin/coach view; role auth is not documented.	FR-3 External interfaces Gemini APIs support workout generation, plan updates and nutrition tips.	FR-4 Transactions FastAPI receives data, calls Gemini, returns results and stores records.
FR-5 Reporting Admin view lists user details and original/updated plans.	FR-6 Business rules Goal/intensity guide generation; feedback revises the original plan.	FR-7 Compliance No specific compliance standard is documented in the source.	FR-8 Other FastAPI /docs supports API testing; Jinja2 renders dynamic results.

Non-Functional Requirements (NFR)

NFR-1 Performance Responsive local interaction; no formal benchmark recorded.	NFR-2 Scalability Modular design supports future managed database and hosting.	NFR-3 Security Keep the Gemini API key in environment variables.
NFR-4 Reliability Local service is available while Uvicorn is running.	NFR-5 Usability Clear responsive HTML/Jinja2 inputs and result sections.	NFR-6 Maintainability Separate AI, routes, database, templates and tests.